Question 1

a) $S = \{E, S, P, D\}$

b) i) $A = \{E, D\}$

$B = \{S\}$

$A \cup B = \{E, D, S\}$

ii) $A \cap B = \{\}$

c) i) $P(E) = P(S) = P(P) = P(D) = 0.250$

$P(A) = 0.250 + 0.250$

$= 0.500$ ※

ii) $P(B) = 0.250$

$P(B') = 1 - 0.250$

$= 0.750$ ※

Question 3

a) $X = \{0, 1, 2, 3, 4\}$

b) $X \sim B(4, 0.20)$

c) i) $P(X=2) = {}^4C_2 (0.2)^2 (0.8)^2$

$= 0.154$

ii) $P(X \leq 1) = P(X=0) + P(X=1)$

$= {}^4C_0 (0.2)^0 (0.8)^4 + {}^4C_1 (0.2)^1 (0.8)^3$

$= 0.819$

d) mean $= (4)(0.2)$

$= 0.8$

variance $= (4)(0.2)(0.8)$

$= 0.64$

Question 4

a, $Y = \{1.2, 3, 4, 5 \ldots\}$ ; all integer variable

b, $Y \sim Geo(p)$

c, $P(Y=5) = (1-0.10)^{5-1}(0.10)$

i) $= 0.066$

ii) $P(Y \leqslant 4) = 1 - P(Y>4)$

$= 1-(1-0.10)^4$

$= 0.344$

Question 5

$\mu = 40$ minutes

$\sigma = 6$ minutes ; $z = \frac{x - \mu}{\sigma}$

a)(i) $P(x < 46)$

$z = \frac{46 - 40}{6} = 1.00$

$P(z < 1.00) = 0.8413$

$P(x < 46) = \underline{0.8413}$ *

(ii) $P(34 < x < 52)$

$x = 34$;

$z = \frac{34 - 40}{6}$

$= -1.00$

$x = 52$

$z = \frac{52 - 40}{6} = 2.00$

$P(z < -1.00) = 0.1587$  $P(z < 2.00) = 0.9772$

$P(34 < x < 52) = 0.9772 - 0.1587$

$= \underline{0.8185}$ *

b) $P(x > x) = 0.05$

$P(x < x) = 0.95$

z-table $(0.95) = 1.645$

$x = z\sigma + \mu$

$= (1.645)(6) + 40$

$= \underline{49.87}$ *

Question 6

$p = 0.6$ ; $q = 0.4$ ; $r = 3$

(a) $X \sim NB(r = 3, p = 0.60)$

(b) $\sigma = \sqrt{\frac{rq}{p^2}} = \sqrt{\frac{3(0.4)}{(0.6)^2}}$

$= \underline{1.826}$ *

(c) negative binomial

$P(X = x) = \binom{x-1}{r-1} p^r q^{x-r}$

$P(x = 5) = \binom{5-1}{3-1} (0.60)^3 (0.40)^{5-3}$

$= \underline{0.20736}$ *